Please write clearly in block capitals.

Centre number | | | | | | Candidate number | | | | |

Surname _____

Forename(s) _____

Candidate signature _____

I declare this is my own work.

Functional Skills Level 1
MATHEMATICS

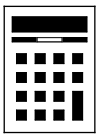
Paper 2 Calculator

Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- a calculator
- mathematical instruments.



For Examiner's Use	
Question	Mark
1–7	
8	
9	
10	
11	
TOTAL	

Instructions

- Use black ink or black ball-point pen. Draw diagrams in pencil.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.
- State the units of your answer where appropriate.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 60.
- You may ask for more answer paper, graph paper and tracing paper. These must be tagged securely to this answer book.

Advice

In all calculations, show clearly how you work out your answer.

Do not write outside the box

Section A

Answer **all** questions in the spaces provided.

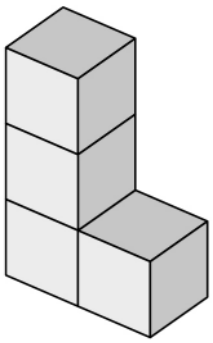
1 Write these numbers in order of size, starting with the **smallest**.

[1 mark]

67 492 351 968 472 410 92 836

Answer _____ , _____ , _____ , _____

2 This shape is made using four cubes.



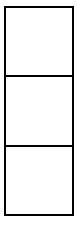
Circle the letter of the plan view.

[1 mark]

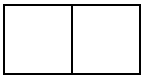
A



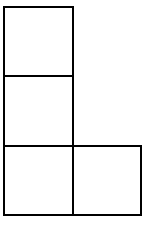
B



C



D



Do not write outside the box

3 Round 17.834 to 2 decimal places.

[1 mark]

Answer _____

4 Circle the fraction equal to 0.75

[1 mark]

$\frac{1}{4}$ $\frac{3}{4}$ $\frac{5}{7}$ $\frac{7}{5}$

5 Work out $24^2 + 37^2$

[2 marks]

Answer _____

Turn over ►

Do not write outside the box

6 Increase 78 by 35%

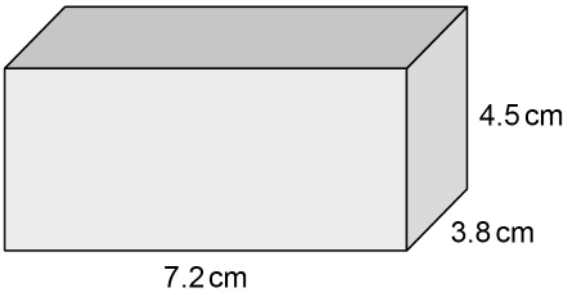
[3 marks]

Answer _____

7 Work out the volume of this cuboid.

State the units of your answer.

[3 marks]



Answer _____

12

Do not write outside the box

Section B

Answer **all** questions in the spaces provided.

8 **School trips**

Mr Nagah and Miss Burns are geography teachers.

They are planning a trip for 34 students to a field centre.

8 (a) Mr Nagah and Miss Burns will go on the trip.

There must be at least 1 teacher for every 8 students on the trip.

How many **more** teachers are needed?

You **must** show your working.

[3 marks]

Answer _____

Turn over ►

Do not write outside the box

8 (b) Mr Nagah is working out the time they should leave school for the field centre.
He makes these notes.

> It will take one and a half hours to get to the field centre
> The students will need 20 minutes to unpack
> We will then have 45 minutes for lunch
> We must be ready for our first activity at 1 pm

Work out the latest time they can set off.

[4 marks]

Answer _____

Do not write outside the box

8 (c) The total cost of the trip is £3265

The geography department pays 20% of this cost.

The rest of the cost is shared equally by the 34 students.

Work out how much each student has to pay.

Give your answer to the nearest pound.

[5 marks]

Answer £ _____

12

Turn over for the next question

Do not write outside the box

9 **Garden Birds**

Simon has a bird feeder in his garden.



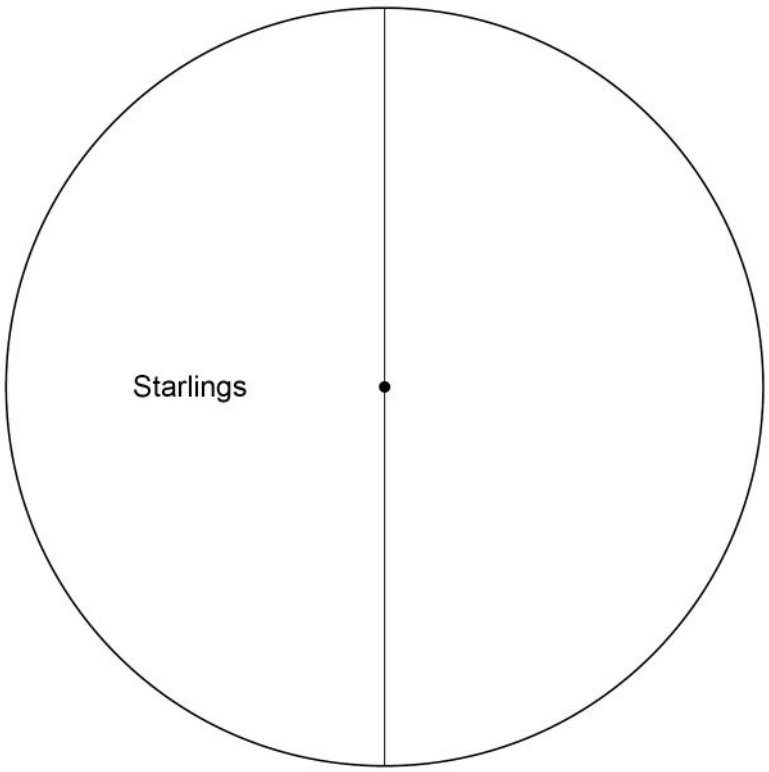
9 (a) On Monday Simon records the types of birds that visit his bird feeder.

He uses this table to work out the angles needed for a pie chart.

	Number	**Angle (degrees)**
Starlings	18	
Sparrows	9	
Blackbirds	6	
Others	3	
	Total = 36	

Complete the pie chart.

[4 marks]



Do not write outside the box

9 (b) Simon takes a photo of each of the 36 birds that visit the bird feeder on Monday.

He decides to choose one of the photos at random.

Simon says,

"The probability that the photo is of a sparrow is **more than** $\frac{1}{3}$ "

Is Simon correct?

You **must** show your working.

[2 marks]

Question 9 continues on the next page

Do not write outside the box

9 (c) Simon counts the birds he sees at the feeder each day for the rest of the week.

Day	Mon	Tues	Wed	Thurs	Fri	Sat	Sun
Number of birds	36	25	30	42	44	26	56

In the same week last year, the mean number of birds per day was 39

This year, has the mean increased, decreased or stayed the same?
Tick (✓) **one** box.

☐ increased ☐ decreased ☐ stayed the same

Show working to support your answer.

[4 marks]

Do not write outside the box

9 (d) Each day, Simon puts 125 **grams** of bird seed in the bird feeder.

He says,

"A 2.8 **kilogram** bag of seed will last for **more than** 3 weeks."

Is Simon correct?

You **must** show your working.

[4 marks]

14

Turn over for the next question

Do not write outside the box

10 **Photographer**

Liam is a photographer.

10 (a) Liam is asked to take photos at Molly's party.

He uses this formula to work out his fee.

Fee = £40 + number of photos × £3.75

Molly wants 18 photos.

She gives Liam six £20 notes.

How much change should she get?

[4 marks]

Answer £ _____

Do not write outside the box

10 (b) Liam also takes school photos.

He sells the photos in four different pack sizes, A, B, C and D.

Liam kept a tally of the packs he sold one day.

Pack	Tally	Frequency
A	𝍸 𝍸 𝍸 𝍸 ||	
B	𝍸 ||||	
C	𝍸 |	
D	𝍸 𝍸 ||||	

Liam says,

"I sold **more than** four times as many of pack A as I did of pack C."

Is he correct?

You **must** show your working.

[3 marks]

Do not write outside the box

10 (c) Liam wants to arrange six photos on a display board in his studio.

The board is a rectangle with width 120 cm and height 140 cm

He has

- one photo of width 60 cm and height 60 cm
- two photos each of width 30 cm and height 40 cm
- three photos each of width 25 cm and height 15 cm

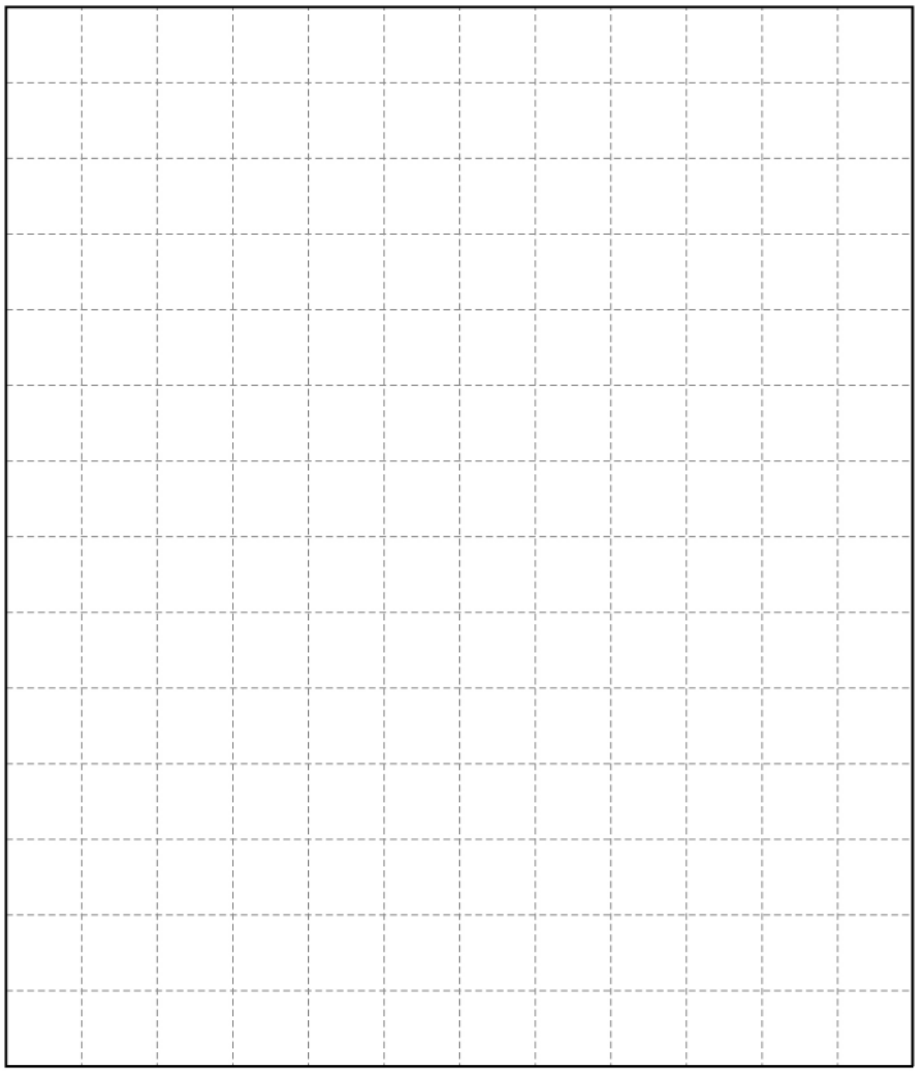
There must be a space of at least 10 cm between photos.

The grid is a scale drawing of the display board.

Use the grid to show how Liam could arrange the photos.

[5 marks]

Scale: represents a 10 cm by 10 cm square



12

Do not write outside the box

11 **Patchwork**

Daisy is making a cushion.

The front of the cushion has a patchwork design made up of 16 square patches.

The back of the cushion is a single piece of fabric.

11 (a) Daisy starts to make a sketch of her design.

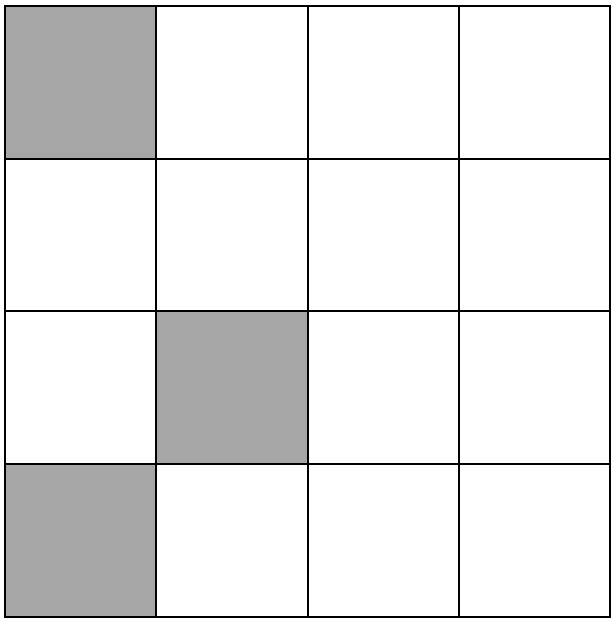
6 of the squares will be grey.

The rest of the squares will be white.

She has shaded 3 of the 6 grey squares.

Shade **3 more** squares so that the design has **at least** one line of symmetry.

[2 marks]



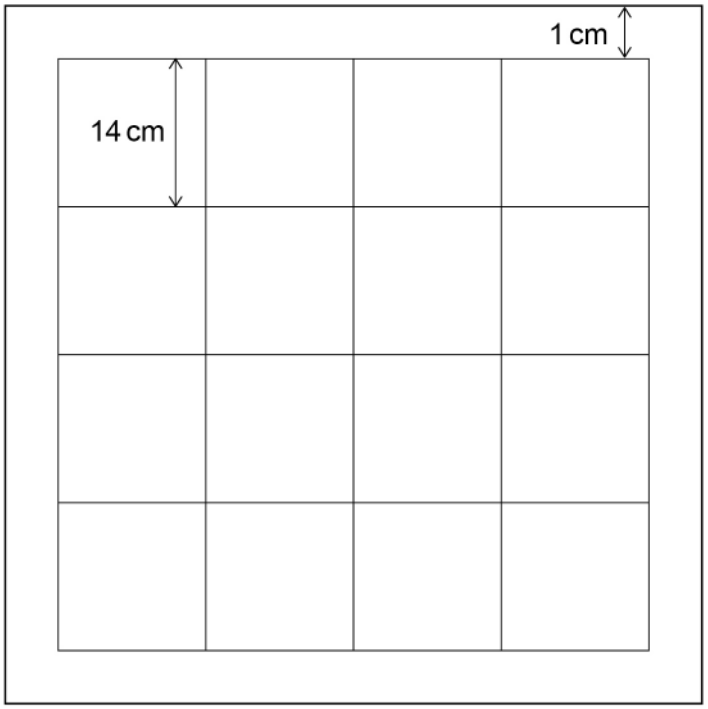
Question 11 continues on the next page

Do not write outside the box

11 (b) Each of Daisy's square patches measures 14 cm by 14 cm

There is a 1 cm border around the patchwork.

The diagram shows the front of Daisy's cushion.



Not drawn accurately

The front and back of the cushion are the same size.

Daisy has a **square** piece of fabric with an area of 3600 cm^2

She wants to cut the back for her cushion from this piece of fabric.

Is this piece of fabric big enough?

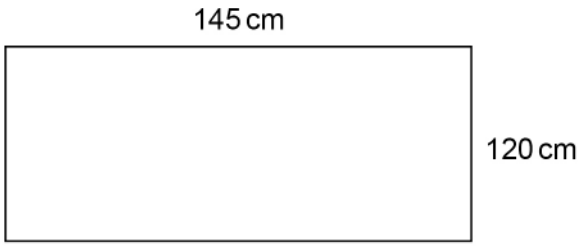
You **must** show your working.

[4 marks]

Do not write outside the box

11 (c) Daisy also sews a patchwork quilt.

The quilt is a rectangle.



Not drawn accurately

Daisy wants to put cord around the outside edges of the quilt.

She has 3.6 **metres** of cord.

How much **more** cord does Daisy need?

State the units of your answer.

[4 marks]

Answer _____

10

END OF QUESTIONS

Do not write outside the box

There are no questions printed on this page

DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED

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Question number	**Additional page, if required.** **Write the question numbers in the left-hand margin.**

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Question number	**Additional page, if required.** **Write the question numbers in the left-hand margin.**

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